PHYS 143 – Mechanics and Thermal Physics

Orientation

How does this course work, and what must you do this week to be ready for the physics?

Hope Donglo

Group E • Office 24, Department of Physics, University of Ghana

Part 1 of 2: Orientation and the course outline (Tuesday 27 October)

First semester 2026/2027 • BSc Level 100 • 3 credits

GROUP E FACILITATOR
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3 credits

8 groups

None prerequisite

None teaching assistants

Quick poll: Who is already logged in to Sakai?

All information and course contents are found in Sakai: a must visit site

Open PHYS 143 on Sakai: sakai.ug.edu.gh

What this course is about

Understand how the physical world works and learn the mathematical tools needed to predict it.

Understanding nature
Mechanics: motion, forces, energy and momentum
Thermal physics: temperature, heat and thermodynamic processes

Mathematical tools
Vectors and calculus for physical modelling

Prediction
Predict and solve real physical problems

Physical world	Mathematical tools
Why cars stop	Vectors
Why projectiles curve	Calculus
	Problem solving
Why bridges expand	
	Quantitative reasoning
How engines produce work	

EACH WEEK Read → Attend → Practice

By the end of this contact you can

1. **Name** the reference text and where slides, notes and videos are posted
2. **Log in** to Sakai and **select a group** with no clash
3. **State** how you are assessed, and the homework and quiz rules
4. **State** when and where Group E meets, and the office hours

What you will study

Now, what does the semester cover?

PHYS 143 is a 3-credit, calculus-based introduction to mechanics and thermal physics for Level 100 science students.

- **Mechanics:** vectors; motion in one, two and three dimensions; Newton's laws; work, energy and power; momentum, impulse and collisions; rotation and gravitation.
- **Thermal physics:** temperature and heat; thermal properties of matter and the kinetic theory of gases; the first and second laws of thermodynamics.
- **Purpose:** to train you to describe physical phenomena, model them mathematically, and predict what happens from physical laws. For the first time in your science career you will solve problems with vectors and calculus, and build the confidence to tackle problems that at first seem difficult.
- **Relevance:** these ideas underpin later physics courses and are used across the physical and medical sciences and engineering.

Discover the outcomes

Before I show you the outcomes, try to work them out from one everyday event.

A car brakes to a stop on a wet road. How far does it travel?

- **What do you need to describe it?** Quantities with units and direction: speed, time, distance, force.
- **How would you model it?** With equations: vectors for direction, calculus for changing speed.
- **How would you predict the stopping distance?** With physical laws: Newton's second law and the work-energy relation.

The abilities this course builds

Describe physical phenomena, model them mathematically, and predict from physical laws.

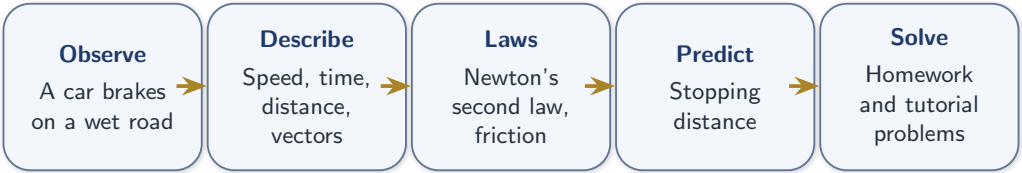
Learning outcomes of the course

Here are the same abilities in the course's own words, with where each is tested.

No.	You will be able to	Tested in
1	**describe** motion with vectors and units, and **model** it with calculus	homework, interim, final
2	**apply** Newton's laws to **predict** the motion of a system	homework, interim, final
3	**analyze** systems with work, energy and momentum conservation, including collisions	homework, interim, final
4	**model** rotation with angular quantities and **explain** Kepler's three laws	Homework 6, final
5	**explain** and **predict** thermal processes with the ideal gas, kinetic theory and the first and second laws	Homework 7, final
6	**evaluate** problems independently from textbooks and online sources, and **apply** physics to other sciences	homework: reading, work on time

How every topic is built

Every chapter follows the same path. Here it is with the braking car.



So where does the learning happen? In class and on Sakai.

Face to face

- Two lecture periods a week in your group, with worked examples and short quizzes.
- Tutorials to discuss the course material and homework, at times and venues set by your group facilitator.

Online on Sakai

- Lecture videos, slides and notes, uploaded every week.
- Homework problem sets and the interim assessment.
- Announcements, the discussion forum and your marks.

Other online methods, such as live sessions on Microsoft Teams or Zoom, are used when a face-to-face meeting cannot hold.

Your weekly routine and ground rules

A steady weekly routine makes this work. Here it is, with the ground rules.

Each week you will

1. Do the assigned reading in Young and Freedman before class.
2. Attend your group's lectures and tutorial, where short quizzes may be given.
3. Work through the homework on Sakai and submit it by its due date.

Ground rules

- Attend lectures and tutorials in your own group.
- Submit homework on the day it is due; late homework is not recorded.
- Quizzes cannot be made up, so come to class prepared.
- Check Sakai every week for materials, announcements and deadlines.

Course map and chapters

Week	Lecture	Topic	Chapter and sections to read	Notes
Week 0 Introduction	none	Orientation; physical quantities	Chapter 1: sections 1.1 and 1.3 to 1.6	Add/drop closes 6 Nov
Week 1	1	Vectors	Chapter 1: sections 1.7 to 1.10	none
Week 2	2	Motion in one, two and three dimensions	Chapters 2 and 3: whole	Homework 1 sent
Week 3	2	Motion, continued	Chapters 2 and 3: whole	Homework 2 sent
Week 4	3	Newton's laws and applications	Chapter 4: whole. Chapter 5: sections 5.1 to 5.4	Homework 3 sent
Week 5	4	Work and kinetic energy	Chapter 6: whole	Homework 4 sent
Week 6	5	Potential energy and conservation	Chapter 7: whole	Homework 5 sent

Interim assessment: Saturday 5 December 2026, on Sakai, Weeks 1 to 5 topics. The University interim period opens 23 November.

Course map and chapters

Week	Lecture	Topic	Chapter and sections to read	Notes
Week 7	7	Momentum, impulse, collisions	Chapter 8: sections 8.1 to 8.5	Reading assignment starts
Week 8	9	Temperature and heat	Chapter 17: whole	Reading continues
Week 9	10	Thermal properties of matter	Chapter 18 except 18.4; also 17.5	Homework 6 sent
Christmas break: 24 December to 3 January				
Week 10	11	First law of thermodynamics	Chapter 19: whole; also 18.4	Homework 7 sent
Week 11	12	Second law of thermodynamics	Chapter 20: whole	Teaching ends 15 Jan
Week 12	none	Revision and tutorials	none	Course evaluation closes 22 Jan
Week 13	none	Final examination	All content	Date set by Academic Affairs

Lecture	Topic	Chapter and sections to read
8	Rotation of rigid bodies; dynamics of rotation; Kepler's laws	Chapter 9: whole. Chapter 10: sections 10.1 to 10.6. Chapter 13: Kepler's laws and gravitation

Not taught in class.
Notes are on Sakai. Tutorials take your questions.

Homework 6 checks this reading, and the material is examined.

REQUIRED

Young and Freedman

University Physics, 15th edition

Chapters 1 to 8, 9, 10, 13, 17 to 20

Copies: Physics Library and Balme Library

Additional reading

Halliday, Resnick and Walker

Tipler

Serway and Jewett

Giancoli

Backhouse and Houldsworth: mathematics

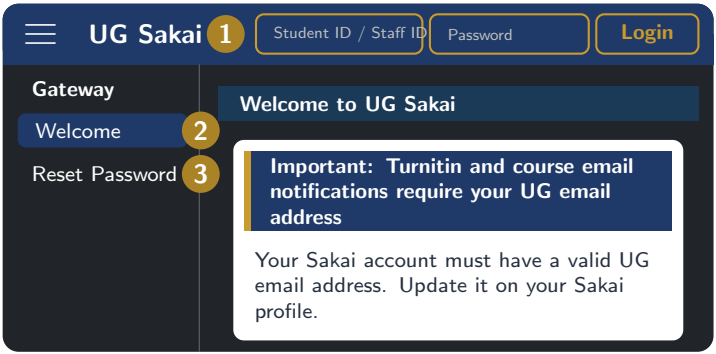
Bring to class: a scientific calculator and a notebook for worked examples

Free resources and simulations

PhET simulation	Topic
Projectile Motion	Motion in two dimensions
Energy Skate Park	Work and energy
Collision Lab	Momentum and collisions
Gravity and Orbits	Gravitation (reading)
Gas Properties	Gases and heat

OpenStax University Physics
Free online and as PDF. Volume 1: mechanics. Volume 2: thermodynamics.

Sakai: logging in



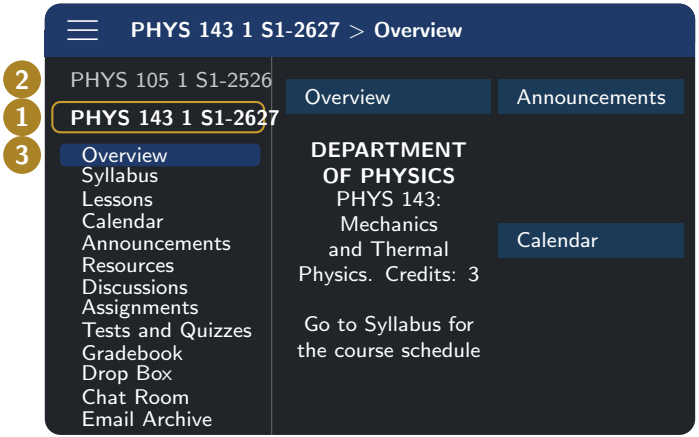
1 Log in with your Student ID and password

2 Read the Welcome notice

3 Forgot your password? Use Reset Password

Add your UG email address to your Sakai profile, or you may miss course emails and Turnitin may fail

Sakai: finding PHYS 143



1 Open **PHYS 143 1 S1-2627** in your site list

2 PHYS 105 1 S1-2526 is a separate site, online this week. Check it too.

3 Start on Overview. The Syllabus has the schedule.

Sakai tour: the left menu

13 tools

Tool	Use
Overview	Course home page
Syllabus	Outline and schedule
Lessons	Weekly content
Calendar	Class and deadline dates
Announcements	Schedule changes
Resources	Videos, slides, notes
Discussions	Ask questions

Tool	Use
Tests and Quizzes	Homework and interim
Gradebook	Your marks
Drop Box	Private files with your facilitator
Email Archive	Messages to the site email
Assignments	Tasks to submit
Chat Room	Live chat with the class

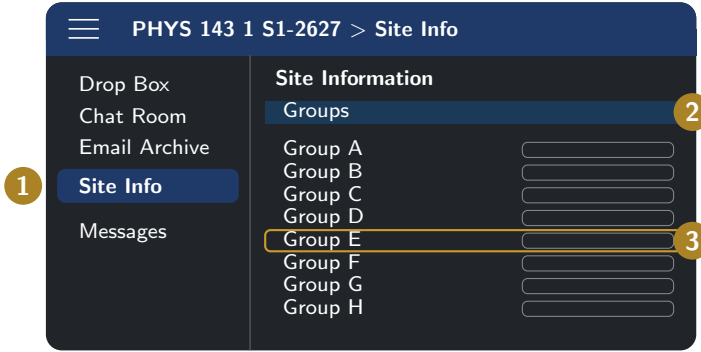
Predict first: which two tools will you open most often? Compare at the end of the semester.

All groups at a glance

Compare all eight timetables before you pick a group, so that you avoid a clash.

Group	Lecture period 1	Lecture period 2
A	Wednesday 14:30 to 15:20, KAB	Thursday 07:30 to 09:20, KAB
B	Monday 11:30 to 13:20, N1	Thursday 07:30 to 09:20, NNB3
C	Monday 15:30 to 17:20, NNB2	Thursday 11:30 to 13:20, NNB2
D	Tuesday 07:30 to 09:20, JQB24	Wednesday 07:30 to 08:20, KAB
E	**Tuesday 09:30 to 11:20, JQB19**	**Wednesday 07:30 to 09:20, N3**
F	Monday 09:30 to 11:20, KAB	Wednesday 08:30 to 09:20, KAB
G	Tuesday 07:30 to 09:20, KAB	Wednesday 15:30 to 16:20, KAB
H	Monday 17:30 to 19:20, L/OT1	Thursday 07:30 to 09:20, N1

KAB: K.A. Busia Building. N1, N3: N-N Block. NNB2, NNB3: New N Block. JQB19, JQB24: Jones Quartey Building. L/OT1: Larway/Oracca Tetteh Building.



1 Open **Site Info** in the left menu

2 Find the groups available to join

3 Join the group that fits your timetable, with no clash. Click **Join**.

Deadline: it will be announced on Sakai

Try not to choose a group that clashes. If every group clashes, there is no solution.

How you are assessed

100%

10%

Homework and quizzes

Seven sets on Sakai, and short quizzes in lectures

30%

Interim assessment

Multiple choice, on Sakai

60%

Final examination

Multiple choice, all content



Homework and quiz rules

Read carefully

Assessment	Rule
Homework	Submit **on the day it is due**. Marked for correct answers with correct units. Due days and times will be announced.
Quizzes	Short questions on reading and recent lectures. The format, length and split of the 10% will be discussed.
Interim, 30%	Make-up only with a doctor's notice. Follow the Sakai announcements for the start time.
Final, 60%	Set by Academic Affairs: date, time and venue
Polls and exit checks	Not marked, but they count toward class participation, which can earn a grade. Follow the Sakai announcements for its weight.

Late homework is not recorded. No make-up quiz is given.

Academic integrity

Academic integrity means honesty, fairness, responsibility, respect and accountability in academic work.

Situation	Rule
Homework and all work you submit	It must be **your own work**
Another student's answers	Copying is **misconduct**
Quizzes, interim, final examination	**No AI tools** unless a facilitator allows
Studying with an AI tool	**Name the tool** and how you used it

Read the **UG Plagiarism Policy** and the **UG Policy on the Use of Artificial Intelligence**

When and where, and getting help

Session	Day and time	Venue
Tuesday lecture	From 27 Oct (orientation), 09:30 to 11:20	JQB19
Wednesday lecture	From 28 Oct (orientation), 07:30 to 09:20	N3
Office hours	Wednesday 10:00 to 12:00. Secondary: yields to the PHYS 105 lab from 18 Nov	Room 24
Tutorial	To be discussed	

If you	Then
Have a question	Email your facilitator: hdonglo@ug.edu.gh
Have a technical problem	Tell your facilitator at once
Have special needs	Tell Dr Joseph Asare this week.

Missed a lecture or a change of schedule? Check **Sakai** first

You know when and where we meet. Here is the rest of the help available.

- **Catch up:** lecture videos, slides and notes are on Sakai every week.
- **Questions between classes:** the Sakai discussion forum, or email your group facilitator.
- **Feedback:** homework and quiz marks are posted on the Sakai Gradebook.
- **Textbooks:** copies of the required text are in the Physics Library and the Balme Library; free open textbooks are on the Resources slide.
- **Online access:** do homework and the interim on a computer with a reliable connection. If a technical problem stops a submission, tell your facilitator straight away.
- **Missed interim for health reasons:** a make-up is possible with a doctor's notice signed by a University physician.
- **Special needs:** tell the Course Coordinator this week; see the UG policy on Students and Staff with Special Needs.

Your roadmap to success

Support only helps if you use it. Here is the weekly plan.

Step	What you do	What it earns	Outcomes
1	Read the assigned chapter before class	you can answer quiz questions on the reading	all
2	Attend lectures and work the examples	quiz marks, and a clear picture of the method	1 to 3
3	Review slides and notes on Sakai	you fill the gaps from class	1 to 5
4	Submit homework on its due day, with units	the 10% for homework and quizzes	2 to 5
5	Bring your hard problems to the tutorial	worked solutions before the tests	2 to 5
6	Ask early: office hours, the Sakai forum, or email	you fix small problems early	all

Your roadmap to success

That is one week. Now the whole semester.

When	Milestone	Outcomes
Week 0, Introduction	join Sakai, choose a group; matriculation Sat 31 Oct; add/drop closes 6 Nov	none
Weeks 1 to 6	Homework 1 to 5, quizzes; University interim period opens 23 Nov	1 to 3
Saturday 5 Dec	interim assessment on Sakai, covering Weeks 1 to 5 (30%)	1 to 3
From Week 7	reading: rotation and Kepler; Homework 6 (Week 9)	4 and 6
24 Dec to 3 Jan	Christmas break	none
Weeks 10 to 12	Homework 7; teaching ends 15 Jan; evaluation 11 to 22 Jan; revision	5
23 Jan to 14 Feb	final examination, multiple choice, all content (60%)	1 to 5

The schedule may change at short notice; lost lectures are made up at times announced on Sakai.

1. **Log in** to Sakai and open PHYS 143 1 S1-2627
2. **Choose your group** before the deadline, with no clash
3. **Read** Chapter 1, sections 1.1 and 1.3 to 1.6, before Wednesday's lecture

Exit check

3 questions, 5 minutes

1. Which assessment is worth 60%, and in what format?

2. What happens to homework submitted after the due day?

3. Where will you find this week's slides and notes?

Answer online on Sakai.

That is the outline. If anything is unclear, this is how to reach me.

GROUP E FACILITATOR

Hope Donglo

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End of Orientation

Next lecture: Physical Quantities and Uncertainties (Wednesday 28 October)

All course materials are posted on Sakai.

Part 2 of 2 – Wednesday 28 October

Physical Quantities and Uncertainties

How do we describe nature with numbers, and how far can we trust them?

Hope Donglo

Group E • Office 24, Department of Physics, University of Ghana

By the end of this lecture you can

That was the orientation. Now the physics begins, with measurement. Here is what you should be able to do by the end.

1. **Convert** quantities to SI units and check that equations are dimensionally consistent.
2. **State** a measurement with its uncertainty and the right significant figures
3. **Round** the result of a calculation correctly
4. **Estimate** a quantity to an order of magnitude

Here is the route for today, so you can see why each step follows from the one before.

1. Measurement
Physics needs **measurement**.

2. Quantities
What we measure are **physical quantities**.

3. Types
Quantities are **fundamental** (independent) or **derived** (defined from fundamental ones).

4. Units
Every quantity needs a **unit**.

5. Standards
Units need **standards**: the SI.

6. Using quantities
Check equations, convert units, and state the uncertainty of every measurement.

We start at the first step: why does physics need measurement at all?

Experimental science

Physics is an **experimental science**.

Observe and find patterns

Physicists **observe** the phenomena of nature and look for **patterns** that relate them.

Laws, principles, theories

Observed regularities are summarised as **laws** or principles, while **theories** provide explanatory frameworks that account for those observations.

Rule: status of a theory

No theory is the ultimate truth. A theory must be testable by experiment, and it remains open to revision when reliable evidence conflicts with its predictions.

Galileo

In the absence of significant air resistance, all objects fall with the same acceleration g. Differences between a feather and a cannonball arise from air drag, not from gravity itself.

Words are not enough

Galileo's claim could be tested only because falling can be measured. Words alone cannot do that.

Science seeks **quantitative** descriptions of nature.

Words depend on the speaker

"The car is fast" and "the rock is heavy". Fast compared with what? Heavy for whom?

Numbers can be checked

"The car travels at $25\ \mathrm{m/s}$" and "the rock has a mass of $2\ \mathrm{kg}$" can be checked by anyone with the right instruments.

Why we measure

A description that cannot be measured cannot be tested, and a theory that cannot be tested is not physics.

Physics needs things that can be measured. What do we call them, and what exactly are they?

Definition: physical quantity

A **physical quantity** is a measurable property of a system or phenomenon that can be expressed by a numerical value and a unit.

Examples

Length, **time**, **mass**, **temperature**, **speed**, **force** and **energy**. They are grouped in a table on the next slides.

Describing you

Weight and height are two physical quantities that describe you.

Not all alike

Some are taken as independent and others are defined from them. That difference is next.

Fundamental and derived quantities

Quantities are not all alike. Some are taken as independent, and others are defined from them.

Fundamental

Taken as independent: it is not defined in terms of other physical quantities. Its meaning is established through a specified measurement procedure (an operational definition), rather than by defining it in terms of other quantities.

Examples: length, time, mass, temperature.

Derived

Defined mathematically in terms of fundamental quantities, even though it may sometimes be measured directly.

Examples: speed, force, energy.

Either way, the quantity is expressed by a numerical value and a unit. The next slide groups the quantities in one table.

Grouping the quantities

Here are the same quantities in one table, grouped by type.

Quantity	Type	How it is specified	SI unit
Length	fundamental	compare with a length standard	metre, m
Time	fundamental	compare with a time standard	second, s
Mass	fundamental	compare with a mass standard	kilogram, kg
Temperature	fundamental	compare with a temperature scale	kelvin, K
Speed	derived	length divided by time	m/s
Force	derived	from mass and acceleration, $F = ma$ (Chapter 4)	newton, N
Energy	derived	from other quantities, for example work = force $\times$ distance (Chapter 6)	joule, J

Each fundamental quantity is fixed by comparison with a standard, and every entry is expressed by a numerical value and a unit. We ask next why the unit matters.

Either kind of quantity is stated as a number, and a bare number tells you nothing.

"3.05"

Writing just "3.05" means nothing: 3.05 what?

"3.05 m"

A basketball hoop is $3.05\ \mathrm{m}$ above the ground.

Reference standard

We compare with a **reference standard**: the hoop is 3.05 times as long as a metre stick, which we define to be $1\ \mathrm{m}$ long. The standard defines the **unit**.

Rule: number and unit

A physical quantity is expressed by a numerical value together with a unit. The number alone is meaningless, so always give the unit.

SI base units

A unit comes from a reference standard, and that standard must be the same for everyone.

Quantity	Unit	Standard
Time	second, s	the time required for $9\,192\,631\,770$ cycles of the microwave radiation of a cesium-133 atom
Length	metre, m	the distance that light travels in vacuum in $1/299\,792\,458$ second
Mass	kilogram, kg	since 2018, based on Planck's constant, $h = 6.62607015 \times 10^{-34}\ \mathrm{kg \cdot m^2/s}$

SI (Système International) is used by scientists and engineers around the world. The gram is 0.001 kg. Kelvin (temperature) and mole (amount of substance) join us in the thermal physics chapters.

Derived units

Speed, force and energy need units too. All of them are built from the base units.

Quantity	Unit	Symbol	In base units
Speed	metre per second	m/s	length divided by time
Force	newton	N	$\mathrm{kg} \cdot \mathrm{m}/\mathrm{s}^2$
Energy	joule	J	$\mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}^2$
Power	watt	W	$\mathrm{J}/\mathrm{s} = \mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}^3$

Other derived units can be formed from the fundamental units: a derived unit is a product or ratio of base units.

Unit prefixes

Some of these units come out very large or very small, so we use prefixes.

Prefix	Symbol	Factor	Young's example
nano	n	10^{-9}	$1\ \mathrm{nm}$: a few times the size of the largest atom
micro	μ	10^{-6}	$1\ \mu\mathrm{m}$: size of some bacteria and other cells
milli	m	10^{-3}	$1\ \mathrm{mm}$: diameter of the point of a ballpoint pen
centi	c	10^{-2}	$1\ \mathrm{cm}$: diameter of your little finger
kilo	k	10^{3}	$1\ \mathrm{km}$: distance in a ten minute walk
mega, giga	M, G	10^{6}, 10^{9}	the same pattern, for example $1\ \mathrm{MW} = 10^{6}\ \mathrm{W}$

A prefix always means the same factor, so $1\ \mathrm{km} = 10^{3}\ \mathrm{m}$ and $1\ \mathrm{kg} = 10^{3}\ \mathrm{g}$.

Using units in equations

With units and standards agreed, we can use quantities in calculations. First, does an equation make sense?

Rule

Each algebraic symbol denotes both a number and a unit. Two terms may be added or equated only if they have the **same units**.

Consistent

$d = vt$, since $10\ \mathrm{m} = \left(2\ \dfrac{\mathrm{m}}{\mathrm{s}}\right)(5\ \mathrm{s})$.

Not consistent

$d = v + t$, since $\mathrm{m} \neq \mathrm{m/s} + \mathrm{s}$.

Units as symbols

Units are treated like algebraic symbols. The s in the denominator cancels, so the product has units of metres, as it must.

Habit

Carry the units through every calculation. A wrong unit means a mistake.

What if the units in a problem do not match? We convert them.

Idea

$1\ \text{min} = 60\ \text{s}$ does not say $1 = 60$. The ratio $\dfrac{1\ \text{min}}{60\ \text{s}}$ equals **1**, and so does its reciprocal.

Right factor

$$3\ \text{min} = (3\ \text{min})\left(\frac{60\ \text{s}}{1\ \text{min}}\right) = 180\ \text{s}.$$

Wrong factor

$(3\ \text{min})\left(\dfrac{1\ \text{min}}{60\ \text{s}}\right) = \frac{1}{20}\ \text{min}^2/\text{s}$, which is not a time.

Evaluate

$180\ \text{s}$ is reasonable, because the second is a smaller unit than the minute, so there are more seconds.

Habit

Work in metres, kilograms and seconds, and convert the final answer last.

Let us apply the method to a real speed.

The world land speed record is $1228.0\ \mathrm{km/h}$ (Thrust SSC). Express it in metres per second.

Predict first: will the number in m/s be larger or smaller than 1228.0?

IDENTIFY Convert km/h to m/s. We need $1\ \mathrm{km} = 1000\ \mathrm{m}$ and $1\ \mathrm{h} = 3600\ \mathrm{s}$.

SET UP
$$v = 1228.0\ \frac{\mathrm{km}}{\mathrm{h}} \times \frac{1000\ \mathrm{m}}{1\ \mathrm{km}} \times \frac{1\ \mathrm{h}}{3600\ \mathrm{s}}$$

EXECUTE
$$v = \frac{1228.0 \times 1000\ \mathrm{m}}{3600\ \mathrm{s}} = \boxed{341.11\ \mathrm{m/s}}$$

EVALUATE Smaller, as predicted. Divide km/h by 3.6 to get m/s: $90\ \mathrm{km/h} = 25\ \mathrm{m/s}$.

Check by proportion

$$1\ \frac{\mathrm{km}}{\mathrm{h}} = \frac{1000\ \mathrm{m}}{3600\ \mathrm{s}}$$

$$1228.0\ \frac{\mathrm{km}}{\mathrm{h}} = X$$

$$X = \frac{1228.0 \times 1000\ \mathrm{m}}{3600\ \mathrm{s}} = 341.11\ \mathrm{m/s}$$

Example 2: converting volume units

A volume needs more care, because the length factor is cubed.

A cut diamond has volume $30.2\ \text{cm}^3$. Express it in mm^3 and in m^3.
Predict first: will the mm^3 number be 10 or 1000 times larger?

IDENTIFY Convert cm^3 to mm^3 and to m^3.

SET UP $1\ \text{cm} = 10\ \text{mm}$ and $1\ \text{m} = 100\ \text{cm}$. Cube each factor.

EXECUTE
$30.2\,\text{cm}^3\,(10\,\text{mm/cm})^3 = 30.2 \times 10^3\,\text{mm}^3$
$\boxed{30\,200\ \text{mm}^3}$
$30.2\,\text{cm}^3\,(1\,\text{m}/100\,\text{cm})^3 = 30.2 \times 10^{-6}\,\text{m}^3$
$\boxed{3.02 \times 10^{-5}\ \text{m}^3}$

EVALUATE 1000 times, as predicted. The m^3 is a huge unit, so its number is tiny.

Check by proportion

$$\frac{1\ \text{cm}^3}{30.2\ \text{cm}^3} = \frac{10^3\ \text{mm}^3}{X}$$

$X = 30.2 \times 10^3\ \text{mm}^3 = 30\,200\ \text{mm}^3$
In m^3: $30.2 \times 10^{-6} = 3.02 \times 10^{-5}\ \text{m}^3$

Uncertainty in a measurement

Every measurement is made against a standard, and none is exact. So how far can we trust one?

Ruler

A ruler is reliable to the nearest millimetre: the book cover is $3\ \mathrm{mm}$, not $3.00\ \mathrm{mm}$.

Micrometer caliper

A micrometer caliper is reliable to the nearest $0.01\ \mathrm{mm}$: the cover is $2.91\ \mathrm{mm}$.

The caliper has the smaller **uncertainty**.

Definition: measurement uncertainty

Every measurement has limited precision. Measurement uncertainty is an estimate of the range within which the true value is expected to lie. It depends on the measurement technique.

The uncertainty is an estimate. How does it differ from the error?

Error

The difference between a measured value and the true value: error = measured value − true value.

Example: true length $10.00\ \text{cm}$, measured $10.12\ \text{cm}$, so the error is $0.12\ \text{cm}$.

We usually do not know the true value, so we cannot know the error exactly. An error is not a mistake.

Uncertainty

An estimate of the range within which the true value is expected to lie.

Example: $10.12 \pm 0.02\ \text{cm}$ means the true value is expected between 10.10 and $10.14\ \text{cm}$.

We never need the true value to state it, so the uncertainty is what we report.

Error is the unknown difference between the measured value and the true value. Uncertainty is our estimate of how large that difference might be.

Every measurement has an uncertainty. How do we write it down?

Plus or minus

56.47 ± 0.02 mm: the true value is likely between 56.45 and 56.49 mm.

Fractional and percent uncertainty

$\frac{0.02 \text{ mm}}{56.47 \text{ mm}} \approx 0.0004$. Percent uncertainty $\approx 0.04\%$.

Percentage given

$47\ \Omega \pm 10\%$: about $\pm 5\ \Omega$, so between 42 and $52\ \Omega$.

Shorthand

$1.6454(21)$ means 1.6454 ± 0.0021. The digits in brackets are the uncertainty in the final digits.

Even small percent uncertainties can be very significant, for example a few metres too far on a journey of hundreds of thousands of metres.

Often the uncertainty is not written at all, and you read it from the digits.

Value	Sig. figs	What it tells you
$2.91\ \mathrm{mm}$	3	the first two digits are known to be correct, the third is uncertain (about $0.01\ \mathrm{mm}$)
$137\ \mathrm{km}$	3	the same number of figures, but the uncertainty is about $1\ \mathrm{km}$
$0.25\ \mathrm{km}$	2	the zero to the left of the decimal point does not count
$0.250\ \mathrm{km}$	3	the zero after the decimal point counts
$1.07\ \mathrm{cm}$	3	a zero between non-zero digits counts

When an explicit uncertainty is not given, the significant figures indicate, by convention, the resolution and precision of the measurement. They are not a rigorous uncertainty: the instrument, the procedure and repeated readings set that.

Zeros at the end of a large number are ambiguous, so we change the notation.

The problem

The distance from the Earth to the Moon is about $384\,000\,000\ \mathrm{m}$. The zeros do not show the number of significant figures.

Scientific notation

Move the decimal point eight places to the left and multiply by 10^8: $3.84 \times 10^8\ \mathrm{m}$, which has three significant figures.

Zeros can count

$4.00 \times 10^8\ \mathrm{m}$ also has three significant figures: the two trailing zeros in 4.00 are significant.

Usual practice: a number between 1 and 10 multiplied by a power of 10.

Now we can write a value clearly. What happens to the digits when we calculate with it?

Rule: multiplication and division

The result has no more significant figures than the factor with the fewest.

$$\frac{0.745 \times 2.2}{3.885} = 0.42 \qquad 3.1416 \times 2.34 \times 0.58 = 4.3$$

Rule: addition and subtraction

The decimal point matters, not the count of figures.

$$123.62 + 8.9 = 132.5 \qquad 27.153 + 138.2 - 11.74 = 153.6$$

123.62 has uncertainty about 0.01, but 8.9 has about 0.1. The sum has uncertainty about 0.1, so write 132.5, not 132.52.

Rounding habits

Three habits keep the final answer reliable.

1. Round, do not truncate

$525\ \mathrm{m}/311\ \mathrm{m} = 1.688\ldots$ to three figures is 1.69, not 1.68.

2. Keep extra digits

Keep extra digits in a multi-step calculation, then round once, at the end.

3. Exact numbers

Exact numbers have no uncertainty: the 2 in $v^2 = v_0^2 + 2a(x - x_0)$ is exactly 2.

In this course

Give answers with three significant figures, or one more at most in doubtful cases. A ten-digit calculator answer overstates the precision.

Let us use the rules in a real calculation.

Find the rest energy E of an electron with $m = 9.11 \times 10^{-31}$ kg.
Predict first: how many significant figures may E have?

IDENTIFY Target: E. Known: m (3 figures) and $c = 2.99792458 \times 10^{8}$ m/s.

SET UP $E = mc^2$, with $1\ \text{J} = 1\ \text{kg} \cdot \text{m}^2/\text{s}^2$.

EXECUTE
$$E = (9.11 \times 10^{-31}\ \text{kg})(2.99792458 \times 10^{8}\ \text{m/s})^2$$
$$= 8.187659678 \times 10^{-14}\ \text{kg} \cdot \text{m}^2/\text{s}^2$$
Round to three figures: $\boxed{E = 8.19 \times 10^{-14}\ \text{J}}$

EVALUATE Three figures, as predicted. A chemical reaction changes 10^{-19} J, so E is about 10^{6} times larger.

Accuracy and precision

Significant figures describe a value. Accuracy and precision describe a measurement, and they differ.

Accuracy

How close a measured value is likely to be to the true value.

Precision

How closely repeated measurements agree with one another. It is often reflected by the number of meaningful digits reported, but a value can have many digits and still be imprecise.

Precise, not accurate

A cheap digital watch shows $10{:}35{:}17$ a.m. and repeats the same reading consistently: very precise. If it runs several minutes slow, it is not accurate.

Accurate, not precise

A grandfather clock with no second hand may be accurate, but its readings agree only to about a minute, so it is not very precise.

A high-quality measurement is both precise and accurate.

Accuracy needs the true value, which we often do not know. Repeating the measurement shows how closely readings agree and gives an estimate of the uncertainty.

Measure the same time four times and look at the spread of the values.

Data

$2.1,\ 2.3,\ 2.2,\ 2.2$ s.

Mean

$\frac{8.8\ \text{s}}{4} = 2.2$ s. The largest deviation from the mean is 0.1 s.

Report

(2.2 ± 0.1) s, an illustrative estimate from the spread of these four readings.

The spread of the repeated readings shows the precision. The maximum deviation is not a universal definition of uncertainty, and systematic effects are not covered; a fuller analysis depends on the method and is treated in PHYS 105.

Sometimes only the order of magnitude is needed.

Even a crude estimate gives useful information. An **order-of-magnitude estimate** aims at the approximate scale of a quantity, usually the nearest power of ten, and is often good only to within a factor of a few.

Guess the data

We know how to calculate the quantity but must guess the data.

Approximate

The exact calculation is too complicated, so we make rough approximations.

Fermi

Enrico Fermi called them **back-of-the-envelope calculations**. Make your best guesses; do not look up a lot of data.

Example

Heartbeats in a lifetime: 70 beats per minute for 70 years is about 3×10^9 beats, so the order of magnitude is 10^9. Using 60 or 80 beats per minute changes the answer by about 15% and does not change the power of ten.

Here is an estimate that tests a well-known story.

Could anyone carry \$1 billion of gold in a suitcase?
Predict first: a few kilograms, or far more?

IDENTIFY Target: the mass of gold worth 10^9 dollars.

SET UP Gold sells for about \$40 per gram, about \$1000 for 25 g, about $\$10^6$ for 25 kg.

EXECUTE 10^9 dollars is 10^3 times more: $10^3 \times 25\ \text{kg} = 25\,000\ \text{kg} = \boxed{25\ \text{tonnes}}$

EVALUATE Far more. No human can lift 25 tonnes. The same mass of water is $25\ \text{m}^3 = 25\,000$ litres, over 100 times a 120 to 160 litre suitcase, so gold would need to be at least 100 times denser than water; it is about 20 times.

Try these three questions.

1. A rock has mass $1.80\ \mathrm{kg}$ and volume $6.0 \times 10^{-4}\ \mathrm{m}^3$. What is its density, with the right significant figures?
   Answer: $\dfrac{1.80\ \mathrm{kg}}{6.0 \times 10^{-4}\ \mathrm{m}^3} = \boxed{3.0 \times 10^3\ \mathrm{kg/m}^3}$; the volume has two figures.

2. Convert $90\ \mathrm{km/h}$ to $\mathrm{m/s}$.
   Answer: $90 \div 3.6 = \boxed{25\ \mathrm{m/s}}$.

3. Estimate the total number of teeth in the mouths of all the students on your campus. Count your own first.
   Answer: (teeth per student) $\times$ (number enrolled). It depends on the enrolment.

Here are the rules in one table.

Idea	Rule
Physical quantity	a measurable property, expressed by a numerical value and a unit
Fundamental, derived	fundamental: independent, set by a measurement procedure; derived: defined from fundamental quantities
SI units	second, metre, kilogram as standards; derived units are products or ratios
Conversion	multiply by unit multipliers equal to 1 and let the units cancel
Consistency	add or equate only quantities with the same units
Uncertainty	state the number $\pm$ uncertainty; if none is given, significant figures indicate resolution by convention
Calculations	multiply or divide: fewest figures; add or subtract: decimal point
Accuracy, precision	close to the true value, versus repeated readings that agree
Estimates	nearest power of ten, often within a factor of a few

Practice questions and fuller derivations are in the tutorial and supplement, posted on Sakai.

That is the end of this lecture. If any part is unclear, this is how to reach me.

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